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Superpermeability: Critical points for applications in fusion

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Abstract

One of key points for the applications of superpermeable membranes in fusion is the question of the upper limit of the permeation flux density. Theoretical estimates give the maximum fluxes of $\sim 10^{19}$ cm^{-2} s^{-1} for the V group metal membranes.

The permeation fluxes of up to 3×10^{16} and 4×10^{17} cm^{-2} s^{-1} were achieved using Nb and thermal atomic hydrogen in the direct membrane experiments and in simulation absorption experiments respectively. Achievement of higher fluxes has been limited by the experimental equipment opportunities.

Another important point is the reliability of membrane operation in the presence of chemically active gases. No negative effects of CH_4, CO and O_2 specially introduced on a superpermeable Nb membrane were detected during 3000 h of operation.

Such membranes coupled with special hydrogen atomizers can be easily incorporated into the up-to-date ITER design concept. The membranes installed in every of the 24 ducts might isolate $> 90\%$ of D/T with the flux densities already reached.

1. Introduction

It was noted in a number of our previous papers (e.g., Refs. [1,2]) that metal membranes may be *superpermeable* to the suprathermal hydrogen particles whose energy (kinetic, internal or chemical) exceeds ~ 1 eV. The term *superpermeability* has been suggested to denote the membrane permeability approaching its conceivable limit: the permeability of an opening of the same area. Superpermeability to thermal hydrogen atoms and fast ions was observed in our experiments on membranes of Ni [2], Fe [3], Pd [1,4] and Nb [1,5,6], as well as by Waelbroeck et al. on Fe [7] acted upon by atomic hydrogen. Hydrogen passing through the superpermeable membranes is simultaneously compressed and purged of all impurities, including helium.

A proposal has been put forward in our previous contributions [8,9] that superpermeable membranes be installed in the pumping ducts of the ITER divertor for the purpose of isolating the major part of the D/T-mixture out of the ITER exhaust, thus substantially reducing the tritium load on the cryogenic pumps serving to pump helium. Non-thermalized hydrogen particles coming from the divertor can be immediately pumped by a membrane placed in the divertor nearest vicinity, while a more distant membrane may pump out thermalized D/T molecules, if they are turned into thermal atoms at a specially introduced atomizer (an incandescent metal surface).

The V Group metals (V, Nb and Ta) appear to be the most suitable materials for the superpermeable membranes. Niobium and tantalum can be also employed for atomizers. This eliminates negative effects from the evaporation of a foreign material onto the membrane. Such an opportunity was successfully demonstrated in our experiments at the MN-1 simulation setup [8,9] with a cylindrical Nb membrane of 600 cm^2 surface area and with Nb and Ta atomizers. The specific pumping speed of 1.5–2 l/s per 1 cm^2 of the membrane has been obtained, which is sufficient for designing an effective membrane system that can be easily placed in the ITER ducts.

Still two essential questions remained relating both

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to the fusion applications and to more fundamental aspects of superpermeability:

(1) What is the upper limit for the permeation flux density at superpermeation, and what limitations does it impose on the applications in fusion?

(2) Will the membrane pumping system be reliable enough in operation, and particularly, in the presence of the expected gas impurities?

The goal of the following is a brief survey of our two-years efforts at finding answers to these two questions.

2. The upper limit for flux density

A superpermeable membrane presents, in fact, a trap for incoming hydrogen [1,10,11]. Suprathermal hydrogen particles (including thermal atoms) can easily pass into the metal, disregarding the high boundary barrier generated by a monoatomic nonmetal film (see Fig. 1). On having entered the metal, the particles "cool down", and atoms equilibrated in the state of solid solution are now effectively prevented by the surface barrier from associative reemission. Due to such a "surface blocking", the dissolved atoms may reach the opposite side of the membrane due to diffusion with a higher probability than be reemitted back, and thus their major part may pass through the membrane, if the outlet barrier is lower than the inlet one (as in Fig. 1).

As the dissolved atoms may overcome the recombination barrier, $\mathscr{E}_r$, only through their association in pairs (a second-order process), the blocking efficiency falls with the flux density rise until superpermeability disappears. The corresponding upper limit of permeation flux density is determined by $\mathscr{E}_r$ (as explicated in our previous contributions [10,11], and as it also follows from the studies by Waelbroeck et al. [7,12] and by Pick and Sonnenberg [13]):

$$j_{max} \propto \frac{1}{L^2} \exp[2(\mathscr{E}_r - \mathscr{E}_b)]/RT. \quad (1)$$

It is this mechanism that usually puts the upper limit to the incident flux density [1]. But this is not true for the V Group metals [1,6] because of the low $\mathscr{E}_b$ level (the lowest among all metals; e.g., it equals about -0.3 eV for Nb and Ta, against $+0.15$ eV for Pd [14]) combined with a rather high $\mathscr{E}_r$ barrier due to nonmetallic surface impurities.

However, the surface barrier suppresses the thermal reemission, but not the reemission induced by energetic hydrogen particles. Thus, superpermeation may be possible, only if the induced reemission is negligible.

E.g., for the case of thermal atoms, the induced reemission (viz., the recombination by the Eley–Rideal mechanism) is negligible when the surface coverage with hydrogen is much less than a monolayer [15]. Therefore, it is the flux giving rise to a monolayer coverage that will be a maximum one:

$$j_{max} \approx 10^{15} \times 10^{13} \exp\left[-\frac{2(\mathscr{E}_r - \mathscr{E}_{ad})}{RT}\right] \text{cm}^{-2}\ \text{s}^{-1}. \quad (2)$$

There are virtually no limitations on the flux magnitude according to Eq. (2) at $T \geq 800$ K (i.e., $j_{max} \geq 10^{20}$ $\text{cm}^{-2}\ \text{s}^{-1}$), if only $(\mathscr{E}_r - \mathscr{E}_{ad}) \leq 0.6$ eV. The latter is usually true, because a monoatomic nonmetal film, if it builds up a $\mathscr{E}_r$ barrier, drastically reduces at the same time the depth of the adsorption well $\mathscr{E}_{ad}$ (see Fig. 1).

Finally, both the flux of diffusion through the membrane, j_{dif}, and the flux of desorption from the outlet boundary, j_{des}^{out}, are limited by a maximum permitted concentration of hydrogen in metal, C_{max}:

$$j_{dif} \leq DC_{max}/L, \quad (3)$$

$$j_{des}^{out} \leq k_r^{out} C_{max}^2. \quad (4)$$

Here D is the diffusion coefficient, L is the membrane thickness, and k_r^{out} is the recombination constant at the outlet surface. For the V Group metals, $D \geq 10^{-4}$ cm^2s^{-1} at $T \geq 800$ K; our experimental data on Nb give $k_r^{out} \geq 10^{24}$ cm^4s^{-1}. Taking, for instance, a maximum tolerable concentration in Nb of 5 at%, we get, with combining the Eqs. (3) and (4), a maximum flux density of $\sim 10^{19}$ $\text{cm}^{-2}\ \text{s}^{-1}$ at $L = 0.3$ mm. This value may apparently be regarded as a theoretical upper limit of flux density, at least for thermal atomic hydrogen.

In the previous studies [1,6,8,9], superpermeability with flux densities of no higher than 5×10^{14} $\text{cm}^{-2}\ \text{s}^{-1}$ was experimentally investigated. We managed to reach a density of about 3×10^{16} $\text{cm}^{-2}\ \text{s}^{-1}$ (Fig. 2) in our

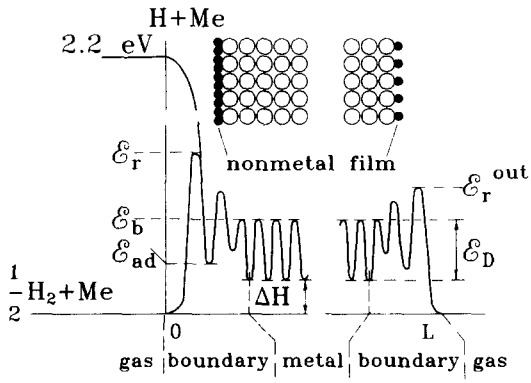


Fig. 1. Potential energy diagram for a hydrogen–metal membrane system. The monatomic nonmetal film generates the $\mathscr{E}_r$ barrier.

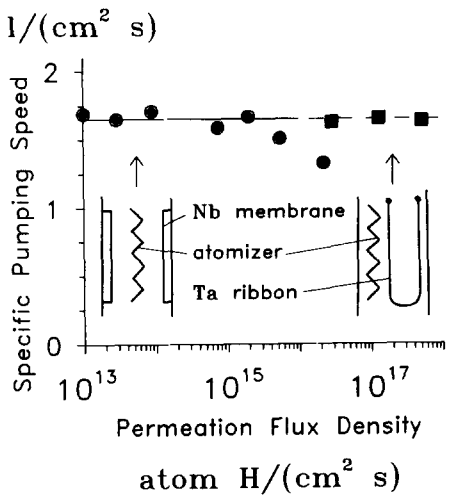


Fig. 2. Rate of hydrogen pumping with superpermeable membrane after atomization of hydrogen at an incandescent Nb (Ta) surface as a function of permeation flux density at the membrane (ribbon) sample temperature of 850–900 K. (●) Direct membrane experiments with a 0.1 mm thick Nb membrane of 600 cm^2 surface area. (■) Evaluation derived from the absorption experiments with a 0.05 mm thick Ta ribbon specimen of 400 cm^2 surface area.

present experiments on a 600 cm^2 Nb membrane combined with Nb and Ta atomizers (with the use of our MN-1 setup [8,9]).

Besides, we arranged simulation adsorption experiments [16] with a 400 cm^2 Ta ribbon specimen (which modeled the membrane) and a Ta atomizer. The adsorption of thermal hydrogen atoms was occurring in the kinetic regime corresponding to superpermeability conditions, i.e., with a probability of absorption in the range of 0.1–0.3 and not depending on temperature, and with a steady state concentration being established with a characteristic time much exceeding the time of diffusion. Such a pattern of behaviour was observed over the whole range of the adsorption fluxes examined, up to 4×10^{17} cm^{-2} s^{-1}. This provides an additional support to our notion that superpermeability must keep up till at least 4×10^{17} cm^{-2} s^{-1}.

A further advancement to even higher flux densities was only arrested by experimental equipment limitations both in the permeation and adsorption experiments.

3. Physico-chemical stability of superpermeable membranes

3.1. General remarks

Primarily, we would like to notice once more that one must not keep the membrane surface clean. Just the other way round, the surface barrier responsible for superpermeability is conditioned by nonmetal impurity films; thus, chemically active gases would rather be expected [1,6,10] to help at the permanent maintenance of a "real" metal surface. The thin surface films hinder at the same time a further accumulation of nonmetallic impurities by effectively inhibiting all chemical reactions at the surface. E.g., a monolayer of O suppresses by 3–5 orders of magnitude the rate of decomposition of CH_4 (Fig. 3) and other hydrocarbons at Ta and Nb (see also a contribution by Hörz et al. [17]).

One might think that a monolayer film is too subtle a thing for practical applications in fusion. It should be remembered however that the surface covered with monoatomic nonmetal layers is, typically, a natural state of thermodynamic equilibrium for a metal surface in vacuum: viz., there are mechanisms which "automatically" maintain a more or less monoatomic coverage through both removing the excessive atoms via their dissolution in bulk metal and replenishing the missing ones via the surface segregation and reactions with ambient gases.

One important empirical fact should be also mentioned in this connection. Many monoatomic nonmetal films are absolutely stable under the action of thermal atomic hydrogen (e.g., under the operational conditions expected for the distant membrane), in spite of its outstanding chemical activity. That has proved being true for the systems O–Ni [2], O–Fe [3], S–Pd [1], C–Pd [1], as well as for O–Nb and O–Ta.

3.2. A 3000 hours' experiment

Fig. 4 demonstrates the results of our long-term experiments in the presence of both the residual gases in our vacuum system (mainly H_2O and CO) and of the chemically active gases purposely admitted during a regular operation of the superpermeable Nb mem-

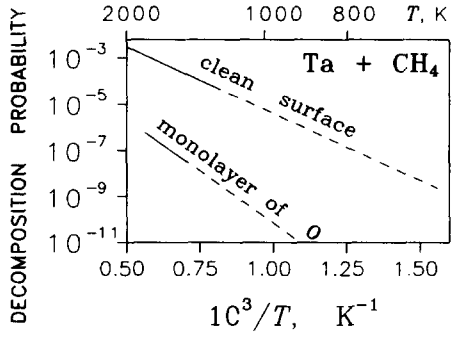


Fig. 3. Effects of an oxygen monolayer on the decomposition of CH_4 molecules on Ta surface. (—) Experimental data by Horz et al. [17]. (– – –) Extrapolation to the range of operational temperatures.

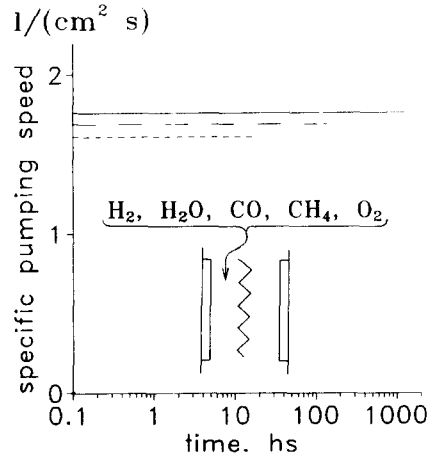


Fig. 4. Effects of residual and specially admitted gases on the rate of hydrogen isolation (pumping) by the Nb membrane (600 cm^2 area. 0.1 mm thickness) with the preliminary atomization of hydrogen at Ta and Nb atomizers. (—) Residual gases ($\approx 5\times10^{-6}$ Pa $H_2O+3\times10^{-5}$ Pa CO). (- - - - -) Admitted CO ($\approx 6\times10^{-4}$ Pa). (– – –) Admitted CH_4 ($\approx 2\times10^{-3}$ Pa). The membrane and atomizer temperatures are 800–950 and 1800–2100 K, respectively.

brane in combination with Ta and Nb atomizers. The pressures of gas impurities corresponded to those that had been expected in the ITER exhaust during the "graphite era". As one can see, H_2O and CH_4 have almost no effect on the system operation, both immediately after the gas admittance initiation and during a long-term (up to 3000 h) operation.

4. Membrane system in the newest ITER concept with gas target divertor [18]

The membrane must pump D/T much faster than the cryopump does (see Fig. 5), if we wish to isolate with it a major portion of tritium, but at the same time, the D/T pressure in the divertor should remain high enough. There are two ways of how the latter may be secured: either through cutting the duct conductivity before the membrane down to a minimum value that would still provide for the required speed of helium evacuation, or through returning a part of D/T evacuated by the membrane back into the divertor.

Let us estimate what characteristics of the membrane system we shall get, if we accept the first way, and if we employ only the *more distant membrane* (combined with an atomizer) whose operation is not controlled by the state of hydrogen near the divertor. We proceed hereby from the specific pumping speed of 16 m^3/s per 1 m^2 of the membrane. We also assume that the pumping speed for He by cryopumps may be varied over the range of 2–200 m^3/s [19], and the fuelling rate may be varied over the range of (0.4–5.4) $\times 10^{22}$ D/T atoms/s [19]. Now, if we occupy with the membranes 1.6 m length in every of the 24 pumping ducts (of 1.5 m in diameter), we get the following membrane system parameters:

(1) *D / T separation efficiency:*
90% at the maximum, and 99.9% at the minimum, cryopump pumping speed;
(2) *Total power consumption:*
0.05–5 MW, depending on the pumping speed by cryopumps;
(3) *Total tritium inventory in the membranes:*
≤ 1.5 g;
(4) *Full pumping speed by the membranes:*
$83 \times 24 = 2000$ m^3/s;
(5) *Permeation flux density:*
$(0.2–3)\times10^{16}$ atom/cm^2/s, depending on the fuelling rate;
(6) *Attainable degree of D / T compression:*
$10^3–10^4$.

Fig. 5 presents an example of the distribution of the D/T concentration and fluxes for the maximum fu-

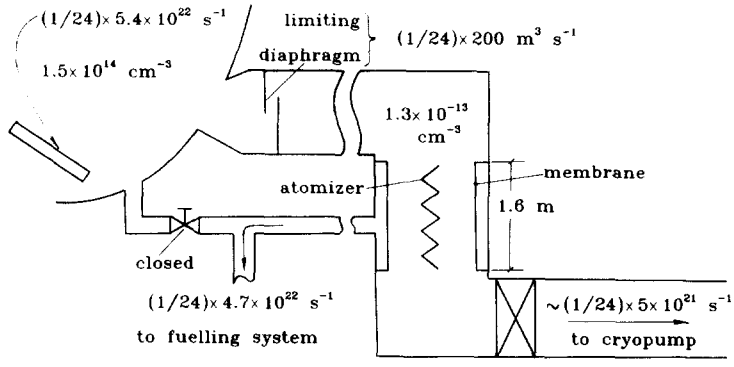


Fig. 5. Distribution of D/T concentrations and fluxes around the ITER exhaust at the maximum fuelling rate (5.4×10^{22} atom/s) and at the maximum pumping speed by cryopumps (200 m^3/s for He), for the case when only a more distant membrane combined with an atomizer is employed to separate D/T from He. The separation efficiency amounts to ~ 90%.

elling rate and the maximum cryopump speed (i.e., for the most stringent operational conditions). One can see that the gas pressure near the membrane is rather low ($\sim 5 \times 10^{-2}$ Pa) even at the maximum fuelling rate, while the flux density does not exceed $5.4 \times 10^{22} \times (24 \times \pi \times 1.5 \times 10^{22} \times 1.6 \times 10^{2}) = 3 \times 10^{16}$ atom/cm^2/s, i.e., a value that has already been reached in our direct membrane experiments.

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